5/22/2020 Online TPT Test Section B

Friday, May 22, 2020 8:20 PM

26. $f(x) = \begin{cases} \frac{1}{4}x^3 - \frac{1}{2}x^2, & x \geq 2 \\ \frac{-6x-6}{x^2+2} & x < 2. \end{cases}$

$f'(x)$ exists at $x=2$?

$f(2) = 2 - 2 = 0$

$f(2^-) = \frac{-18}{6} = -3.$

$\because f(2^+) \neq f(2^-)$

$\therefore$ doesn't exist.

27. $(x+y) = e(x+y)$, what's the $\frac{dy}{dx}$ at point $(\frac{1}{2}, \frac{1}{2})$

$(e-1)y = (1-e)x$

$y = -x, \quad \therefore y' = -1$

28. $A = \begin{pmatrix} ? & ? & ? \\ 0 & 0 & ? \\ 0 & 0 & ? \end{pmatrix}$ each ? is potentially different.

and all ? are real numbers.

A. $\det|A| = 0$

B. $\det|A| \neq 0$ .....

$\det|A| = \begin{vmatrix} a & b & c \\ 0 & 0 & d \\ 0 & 0 & e \end{vmatrix} = a \cdot 0 - b \cdot 0 + c \cdot 0 = 0$

A

29. Area R, x-axis, $y=1$, $2y^2 = 1 - \frac{1}{x^2}$.

$\frac{1}{x^2} = 1 - 2y^2 \quad \therefore x = \pm \frac{1}{\sqrt{1-(\sqrt{2}y)^2}}$

$\sqrt{1-(\sqrt{2}y)^2}$

$$S = 2\int_0^1 \frac{1}{\sqrt{1-(\sqrt{2}y)^2}}\,dy$$

$$= 2\left[\frac{1}{\sqrt{2}}\arcsin(\sqrt{2}y)\right]_0^1$$

...

30. 360% of 360.

$360 \times 360\% = 1296$

31. Which vetor is perpendicular to the plane: $P(1,3,1)$, $Q(2,2,5)$ $R(-1,-2,-1)$.

$\vec{a} = \vec{PQ} = (1,-1,4)$
$\vec{b} = \vec{RP} = (2,5,2)$

$$\vec{C} = \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 4 \\ 2 & 5 & 2 \end{vmatrix} = (-2-20)\vec{i} - (2-8)\vec{j} + (5+2)\vec{k}$$
$$= -22\vec{i} + 6\vec{j} + 7\vec{k}$$

A, $22\vec{i} - 6\vec{j} - 7\vec{k}$
B, ...

32. $x^2+y^2=2xy$.

A. $x=y$
B. $x>y$
C. $x<y$
D. ...

$(x-y)^2=0$, $x=y$.
A.

33. What's the ratio of the volumn of a rectangular solid increase if its length, heighth, width all increase by 25%

$(1.25)^3 - 1 = 0.953125$.

A. 95%

A.

34. $f(x) = \begin{cases} ax^2+1, & x \geq 1 \\ bx+3, & x < 1 \end{cases}$ is differentiable at $x=1$, then $a=?$

① $f(1^+) = f(1^-)$

$a+1 = b+3 \Rightarrow b = a-2$

② $f'(1^+) = f'(1^-)$

$2a = b$

$\rightarrow a = -2$

35. which is the common factor of x^2-4x-5, x^2-6x-7

A. $x-5$

B. $x-7$

C. $x+1$

D. ...

$x^2-4x-5 = (x-5)(x+1)$

$x^2-6x-7 = (x-7)(x+1)$

C.